

Phase 2 — CIC-DDoS2019 Dataset Characterization & Audit

CEP-RF Project | Research Dataset Audit Report

Purpose. Phase 2 establishes a reproducible understanding of the CIC-DDoS2019 dataset before modeling: schema consistency, labels, feature quality, redundancy, leakage screening, taxonomy, and dataset characterization. Raw Parquet files were not modified.

1. Phase 2 in brief

The audit inspected all Parquet files, validated schemas and labels, identified constant and near-constant features, measured redundancy through correlation, screened potential leakage/identifier columns, grouped features into a taxonomy, and generated reproducible CSV reports.

Component	Result
File inventory	17 Parquet files
Schema	78 columns per file; integer-width differences documented
Target	Label; 18 exact labels
Constant features	12 global zero-variance features
Near-constant features	0 pairs
Correlation	33 pairs at $ r \geq 0.95$
Leakage screening	17 keyword matches; all legitimate Protocol false positives
Taxonomy	78 columns, 11 groups
Candidate features	65 variable numeric features

2. Dataset overview

Metric	Result
Parquet files	17
Total flow rows	431,371
Total columns	78
Numeric candidate features	77
Target	Label
Unique labels	18
Benign rows	97,831 (22.6791%)
Attack rows	333,540 (77.3209%)
Variable numeric features after constants	65
Taxonomy groups	11

Class balance: attack traffic is approximately 3.41× benign traffic. Phase 3 should therefore use class-aware evaluation rather than accuracy alone.

Label finding: filenames are not treated as class labels. The actual *Label* column is authoritative, and multiple files contain both benign and attack labels.

3. Schema consistency

All 17 files expose the same 78-column structure. Six columns have different integer storage widths across files; these are numeric-width differences rather than semantic type changes.

Column	Observed types
Bwd Header Length	int16 / int32 / int64
Fwd Act Data Packets	int8 / int16
Fwd Header Length	int16 / int32 / int64
Fwd Seg Size Min	int8 / int32
Subflow Fwd Packets	int16 / int32
Total Fwd Packets	int16 / int32

These features are retained. Raw files remain unchanged; correlation analysis normalizes numeric values only for analysis.

4. Constant-feature analysis

Twelve features are globally constant: each has one unique value (0.0) and zero variance. Such features cannot discriminate between classes and are safe candidates for removal.

#	Feature
1	Bwd PSH Flags
2	Fwd URG Flags
3	Bwd URG Flags
4	FIN Flag Count
5	PSH Flag Count
6	ECE Flag Count
7	Fwd Avg Bytes/Bulk
8	Fwd Avg Packets/Bulk
9	Fwd Avg Bulk Rate
10	Bwd Avg Bytes/Bulk
11	Bwd Avg Packets/Bulk
12	Bwd Avg Bulk Rate

Accounting: 77 numeric candidate features – 12 global constants = **65 variable numeric candidate features**.

5. Near-constant analysis

Per-file variance and unique-value checks found **0 near-constant feature/file pairs** under the configured criteria. No additional feature removal is justified from this test.

6. Correlation and redundancy

Correlation was calculated after excluding the 12 global constants, using $|r| \geq 0.95$ and deterministic memory-safe sampling. The audit found **33 highly correlated pairs**.

Redundancy area	Examples
Packet counts	Total Fwd Packets ↔ Subflow Fwd Packets (1.0); Total Fwd Packets ↔ Fwd Act Data Packets (≈0.9866)
Subflow volume	Fwd Packets Length Total ↔ Subflow Fwd Bytes (1.0); Bwd Packets Length Total ↔ Subflow Bwd Bytes (1.0)
Timing/IAT	Flow Duration ↔ Fwd IAT Total (≈0.9994); Flow IAT Max ↔ Fwd IAT Max (≈0.9986)
Packet size	Fwd Packet Length Mean ↔ Avg Fwd Segment Size (1.0); Packet Length Mean ↔ Avg Packet Size (≈0.9785)
TCP flags	Fwd PSH Flags ↔ RST Flag Count (1.0) in sampled analysis

Decision: these are redundancy candidates, not automatic deletions. They are carried forward for later feature-selection experiments after establishing a baseline.

7. Leakage / identifier audit

Measure	Result
Keyword matches	17
Confirmed leakage	0
Candidates requiring review	0
Allowlisted legitimate matches	17
Matched column	Protocol
Decision	Retain Protocol

All 17 automated matches were Protocol. The audit classifies Protocol as a legitimate network feature and allowlists it. The Label target is not a model feature.

8. Feature taxonomy

All 78 columns were assigned to 11 groups for interpretation and later feature-importance/SHAP analysis. The taxonomy is an interpretability artifact and does not itself remove features.

Group	Typical contents
Timing / IAT	Flow and inter-arrival timing statistics
Packet count	Forward/backward packet counts
Packet size	Min/max/mean/std/variance
Rate / throughput	Bytes/s and packets/s
TCP flags	SYN/ACK/RST and related flags
Subflow volume	Subflow packets and bytes
Window / header	Header lengths and initial windows
Activity / idle	Active and idle interval statistics
Bulk transfer	Bulk-related measurements
Protocol	Network protocol feature
Target	Label

9. Final Phase 2 feature accounting

78 total columns → 1 target (Label) → 77 numeric candidate features → 12 global constants removed → 65 variable numeric candidate features. The 33 correlation pairs remain reported for later optimization; confirmed

leakage is zero and Protocol is retained.

10. Reports generated

#	Report
1	dataset_summary.csv
2	column_summary.csv
3	schema_comparison.csv
4	schema_type_differences.csv
5	label_distribution.csv
6	label_summary.csv
7	label_file_mapping.csv
8	constant_feature_analysis.csv
9	near_constant_features.csv
10	high_correlation_pairs.csv
11	correlation_matrix.csv
12	potential_leakage_columns.csv
13	feature_taxonomy.csv
14	dataset_characterization.csv

Two legacy CSV reports were preserved in the directory for compatibility.

11. Phase 2 conclusions

- Dataset inventory and schema are characterized: 17 files, 431,371 flows, 78 columns.
- Label is established from the data, not filenames.
- 12 global constants are identified for removal.
- No near-constant pairs were found under the audit criteria.
- 33 high-correlation pairs document substantial redundancy but are not automatically removed.
- No confirmed leakage was found; all 17 matches were legitimate Protocol false positives.
- 65 variable numeric candidate features form the Phase 2 starting set.
- Raw data remains unchanged.

12. Transition to Phase 3

Phase 2 is complete from an audit and characterization perspective. Phase 3 can now establish an RF baseline using the 65 variable candidate features. Feature-selection methods should be evaluated against that baseline rather than applied blindly during the audit.

PHASE 2 STATUS: COMPLETE

Authoritative Phase 2 candidate feature count: **65 variable numeric features**.